

Student's name

Student's number

Teacher's name



PLC PRESBYTERIAN
LADIES' COLLEGE
SYDNEY
1888

2014
TRIAL
HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time - 5 minutes
- Working time - 2 hours
- Write using blue or black pen
Black is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total Marks – 70

Section I: Pages 3-6

10 marks

- Attempt questions 1-10, using the answer sheet on page 13.
- Allow about 15 minutes for this section

Section II: Pages 7-10

60 marks

- Attempt questions 11-14, using the booklets provided.
- Allow about 1 hours 45 minutes for this section

Question	1-10	11	12	13	14	Total	%
Marks	/10	/15	/15	/15	/15	/70	

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Section I

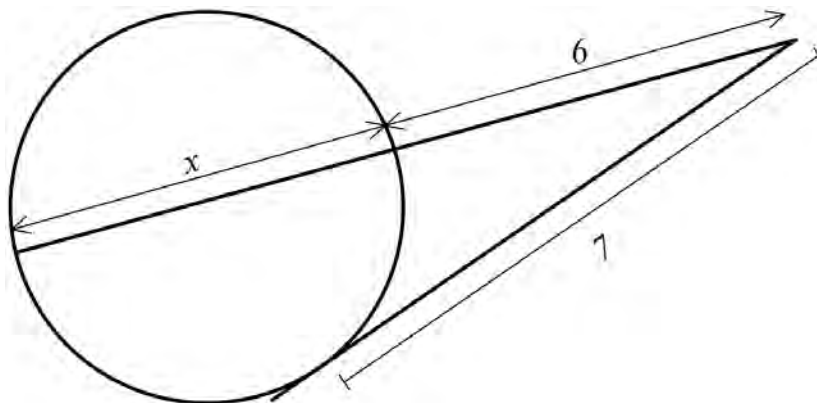
10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10

1.



What is the value of x ?

- (A) 1
- (B) $1\frac{1}{6}$
- (C) $2\frac{1}{6}$
- (D) $8\frac{1}{6}$

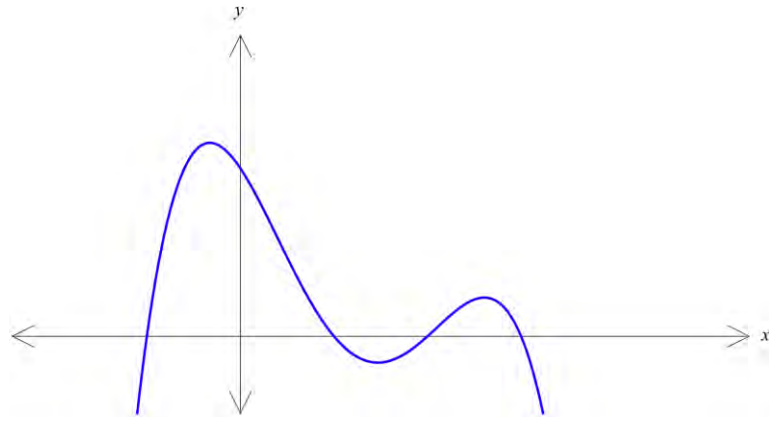
2.

What is the solution of $\frac{5}{1-x} < 3$?

- (A) $x < -\frac{2}{3}, x > 1$
- (B) $x < -\frac{2}{3}$
- (C) $x > 1$
- (D) $-\frac{2}{3} < x < 1$

3. What is the value of $\lim_{x \rightarrow 0} \left(\frac{2x}{\sin 5x} \right)$?
- (A) $\frac{2}{\sin 5}$
- (B) $\frac{2}{5}$
- (C) $\frac{5}{2}$
- (D) $\sin 3x$
4. What are the co-ordinates of the point which divides the interval joining $A(3, -2)$ and $B(-5, 4)$ **externally** in the ratio of 5:3?
- (A) $\left(0, \frac{1}{4} \right)$
- (B) $(15, -11)$
- (C) $(-17, 13)$
- (D) $\left(-2, \frac{3}{2} \right)$
5. The inverse of the function $f(x) = e^{2x-1}$ is?
- (A) $f^{-1}(x) = -e^{2x-1}$
- (B) $f^{-1}(x) = \frac{e^{x+1}}{2}$
- (C) $f^{-1}(x) = -\log_e(2x+1)$
- (D) $f^{-1}(x) = \log_e \sqrt{x} + \frac{1}{2}$

6. The graph below shows a polynomial function, $y = P(x)$.



Which of the following could be the equation of $P(x)$?

- (A) $P(x) = (x+1)(x+2)(x+3)(x-1)$
 (B) $P(x) = -(x+1)(x+2)(x+3)(x-1)$
 (C) $P(x) = (x+1)(x-1)(x-2)(x-3)$
 (D) $P(x) = -(x+1)(x-1)(x-2)(x-3)$
7. The co-efficient of x^2 in the expansion $(2x-3)^5$ is?

- (A) -1080
 (B) -540
 (C) 540
 (D) 1080

8. Using $u = \cos x$,

$\int_0^{\frac{\pi}{3}} \sin^3 x \cos^4 x dx$ can be expressed in terms of u as

- (A) $\int_0^{\frac{\pi}{3}} u^6 - u^4 du$
 (B) $\int_{\frac{1}{2}}^1 u^6 - u^4 du$
 (C) $\int_{\frac{1}{2}}^1 u^4 - u^6 du$
 (D) $\int_0^{\frac{\sqrt{3}}{2}} u^4 - u^6 du$

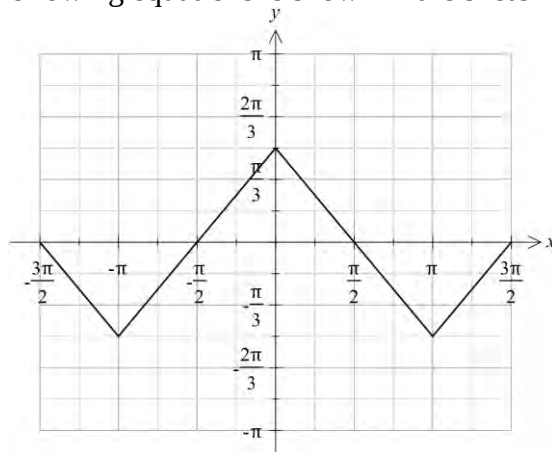
9. A particle is moving along the x -axis, initially moving to the left from the origin. Its velocity and acceleration are given by
 $v^2 = 2 \log_e(3 + \cos x)$ and

$$\ddot{x} = \frac{-\sin x}{3 + \cos x}.$$

Which of the following describes the subsequent motion?

- (A) Moves only to the left, alternately speeding up and slowing down, without becoming stationary.
- (B) Moves only to the left, alternately slowing to a stop and speeding up.
- (C) Slowing to a stop, then heading to the right forever.
- (D) Oscillates between two points.

10. Which of the following equations is shown in the sketch below?



- (A) $y = \cos^{-1}(\sin x)$
- (B) $y = \sin^{-1}(\cos x)$
- (C) $y = \sin^{-1}(x) + \sin(x)$
- (D) $y = \cos^{-1}(x) + \cos(x)$

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section.

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

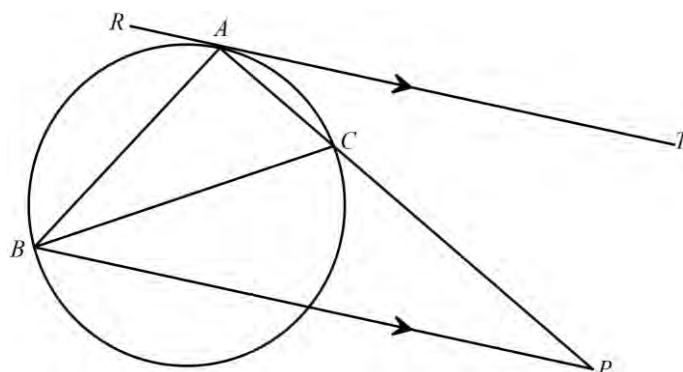
Question 11 (15 marks) Use a SEPARATE writing booklet.

- a) Find the acute angle between the lines $x - 2y + 3 = 0$ and $y = 3x - 1$ at their point of intersection. 2
- b) Find $\int \frac{1}{25 + 9x^2} dx$. 2
- c) Find $\frac{d}{dx} \sin^{-1}(2x^3)$ 2
- d) The polynomial $P(x) = x^3 - 3x^2 + kx + 12$ has 3 roots. It is known that two of the roots are of equal magnitude but opposite in sign. What is the value of k ? 3
- e) Explain why Newton's method does not work for the root of the equation $x^3 - 3x + 6 = 0$ if the initial approximation is chosen to be $x = 1$. Use mathematics to support your answer. 2
- f) If $\cot^2 \theta - \cot \theta = 1$, where $0 < \theta < \frac{\pi}{2}$, 1
- (i) Show that $\cot \theta = \frac{1 + \sqrt{5}}{2}$. 1
- (ii) Hence, show that the exact value of $\cot 2\theta = \frac{1}{2}$. 3

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

- a)** AT is a tangent and is parallel to BP . Prove that $\angle ABP = \angle ACB$. **3**



- b)** A roast duck is taken out of the oven once it is cooked. A thermometer records the temperature of the duck to be 75°C . The roast duck is then allowed to cool in a room with a constant temperature of 23°C .
- (i) Show that $T = 23 + Ae^{-kt}$ satisfies the differential equation **1**

$$\frac{dT}{dt} = -k(T - 23)$$
 where
 T is the temperature of the duck in degrees Celsius, $^{\circ}\text{C}$,
 t is the time in minutes and
 k is a constant.
- (ii) Show that $A = 52$. **1**
- (iii) Find the value of k (in exact form) if after 5 minutes the duck's temperature is 65°C . **2**
- (iv) Bacteria start to develop rapidly in the duck after 8 minutes. What will be the duck's temperature when the bacteria start to develop? Answer to the nearest degree. **1**
- c)** Using the substitution $u = e^x$, find $\int \frac{e^x dx}{\sqrt{1 - e^{2x}}}$ **2**
- d)** (i) Find the domain and range of the function $f(x) = \sin^{-1}(2x)$. **1**
(ii) Sketch the graph of the function $f(x) = \sin^{-1}(2x)$. **1**
(iii) The region bounded by the graph $f(x) = \sin^{-1}(2x)$ and the x -axis between $x = 0$ and $x = \frac{1}{2}$ is rotated about the y -axis to form a solid. **3**
Find the exact volume of the solid.

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- a)** The speed v m/s of a particle moving in a straight line is given by $v^2 = 84 + 16x - 4x^2$ where the displacement of the particle relative to a fixed point is x cm.
- (i) Find an expression for the particle's acceleration in terms of x . **2**
- (ii) Hence show that the particle is moving in simple harmonic motion. **1**
- (iii) Find the period, amplitude and centre of motion. **2**
- b)** (i) The monic polynomial, $P(x)$, has a root at $x = 3$, a double root at $x = -1$ and is of degree 4. If the polynomial passes through the point $(1,0)$, find the equation of the polynomial $P(x)$. **2**
- (ii) The polynomial $Q(x)$ has equation $Q(x) = x^2 + 1$. **2**
Show that $\frac{P(x)}{Q(x)}$ has a remainder of $4x + 8$.
- c)** A balloon has the shape of a right circular cylinder of radius r and length twice the radius, with a hemisphere at each end of radius r . The balloon is being filled at the rate of $10\text{ cm}^3/\text{s}$. Find the rate of change of r when $r = 8$ centimetres **2**
- d)** The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are the ends of a focal chord on the parabola $x^2 = 4ay$.
- (i) Show that PQ has equation $(p + q)x - 2y - 2apq = 0$. **1**
- (ii) Show that $pq = -1$ if PQ is a focal chord. **1**
- (iii) Show that the equation of the tangent at P is $y = px - ap^2$. **1**
- (iv) Hence find the locus of the point of intersection of the tangents at the ends of the focal chord. **1**

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

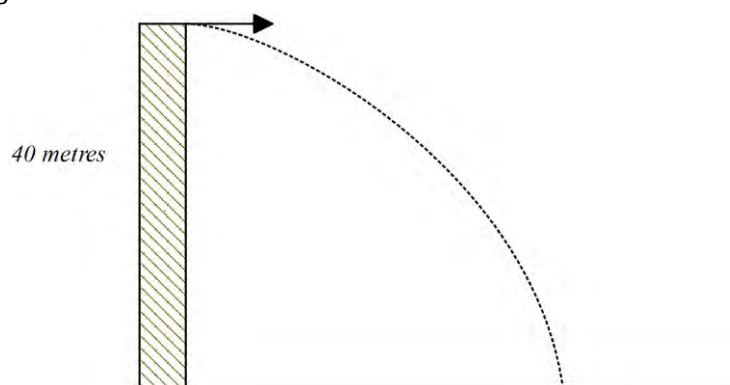
- a) Prove by Mathematical Induction that 3

$$\sum_{r=1}^n \log_e \left(\frac{r+1}{r} \right) = \log_e (n+1) \text{ for all positive integers, } n.$$

- b) Find the general solutions for $2 \cos x = \sqrt{3} \cot x$. 3

- c) An object is projected horizontally from the top edge of a vertical cliff 40 metres above sea level with a velocity of 40 m/s .

Take $g = 10 \text{ m/s}^2$.



- (i) Using the top edge of the cliff as the origin, prove that the parametric equations of the path of the object are: 2

$$x = 40t \qquad y = -5t^2 + 40$$
- (ii) Calculate when and where the object hits the water. 1
- (iii) Find the velocity and angle of the object the instant it hits the water. 2
- d) 4
 By considering $(1-x)^n \left(1 + \frac{1}{x}\right)^n$, or otherwise, express

$$\binom{n}{2} \binom{n}{0} - \binom{n}{3} \binom{n}{1} + \dots + (-1)^n \binom{n}{n} \binom{n}{n-2}$$

 in simplest form.

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

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Mathematics Extension 1:

Multiple Choice Answer Sheet

Student Number _____

Completely fill the response oval representing the most correct answer.

- | | | | | | | | | |
|-----|---|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|
| 1. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 2. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 3. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 4. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 5. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 6. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 7. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 8. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 9. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 10. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |

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Mathematics Extension 1:

Multiple Choice Answer Sheet

Student Number ANSWERS

Completely fill the response oval representing the most correct answer.

- | | | | | |
|-----|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| 1. | A ☐ | B ☐ | C ☒ | D ☐ |
| 2. | A ☒ | B ☐ | C ☐ | D ☐ |
| 3. | A ☐ | B ☒ | C ☐ | D ☐ |
| 4. | A ☐ | B ☐ | C ☒ | D ☐ |
| 5. | A ☐ | B ☐ | C ☐ | D ☒ |
| 6. | A ☐ | B ☐ | C ☐ | D ☒ |
| 7. | A ☒ | B ☐ | C ☐ | D ☐ |
| 8. | A ☐ | B ☐ | C ☒ | D ☐ |
| 9. | A ☒ | B ☐ | C ☐ | D ☐ |
| 10. | A ☐ | B ☒ | C ☐ | D ☐ |

Academic Year		Calendar Year	
Course		Name of task/exam	

Section I

$$1. (x+6)6 = 7^2$$

$$6x + 36 = 49$$

$$6x = 13$$

$$x = \frac{13}{6}$$

$$x = 2\frac{1}{6}$$

$\therefore C$

$$2. \frac{5}{1-x} < 3$$

Critical pts:
 $x = 1$

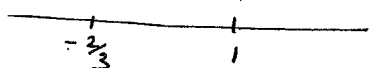
$$\frac{5}{1-x} = 3$$

$$5 = 3(1-x)$$

$$5 = 3 - 3x$$

$$2 = -3x$$

$$x = -\frac{2}{3}$$



check $x = 0$

$$\frac{5}{1} < 3 \quad \text{false}$$

$$\therefore x < -\frac{2}{3}, x > 1$$

$\therefore A$

$$3. \lim_{x \rightarrow 0} \frac{2x}{\sin 5x} = \frac{2}{5}$$

$\therefore B$

$$4. A(3, -2) \quad B(-5, 4)$$

$$-5 : 3$$

$$\therefore \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

$$= \left(\frac{-5(-5) + 3(3)}{-2}, \frac{-5(4) + 3(-2)}{-2} \right)$$

$$= \left(\frac{25+9}{-2}, \frac{-20-6}{-2} \right)$$

$$= (-17, 13)$$

$\therefore C$

$$5. f: f(x) = e^{2x-1}$$

$$f^{-1}: x = e^{2y-1}$$

$$\ln x = (2y-1)$$

$$\ln x + 1 = 2y$$

$$y = \frac{1}{2}(\ln x + 1)$$

$$\therefore f^{-1}(x) = \frac{1}{2} \ln x + \frac{1}{2}$$

$$= \ln x^{\frac{1}{2}} + \frac{1}{2}$$

$$= \ln \sqrt{x} + \frac{1}{2}$$

$\therefore D$

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6 D

$$I \quad (2x-3)^5$$

$$\begin{aligned} T_{k+1} &= {}^nC_k (a)^{n-k} (b)^k \\ &= {}^5C_k (2x)^{5-k} (-3)^k \\ &= {}^5C_k 2^{5-k} (-3)^k x^{5-k} \end{aligned}$$

for coefficient of x^2

$$\therefore 5-k=2$$

$$k=3$$

$$\begin{aligned} \therefore \text{coeff} &= {}^5C_3 2^2 (-3)^3 \\ &= 10 \times 4 \times -27 \\ &= -1080 \end{aligned}$$

 $\therefore A$

$$8 \quad \int_0^{\pi/3} \sin^3 x \cos^4 x \, dx$$

$$u = \cos x$$

$$\text{when } x=0 \quad u=1$$

$$\text{when } x=\frac{\pi}{3} \quad u=\frac{1}{2}$$

 $\therefore$ integral goes from 1 to $\frac{1}{2}$ or - integral goes from $\frac{1}{2}$ to 1

$$\therefore u = \cos x$$

$$du = -\sin x \, dx$$

$$\sin x = \sqrt{1-\cos^2 x}$$

$$= \sqrt{1-u^2}$$

$$\int_0^{\pi/3} \sin^2 x \sin x \cos^4 x \, dx$$

$$= \int_{\frac{1}{2}}^1 (-)(1-u^2) u^4 \, du$$

$$= \int_{\frac{1}{2}}^1 (1-u^2) u^4 \, du$$

$$= \int_{\frac{1}{2}}^1 (u^4 - u^6) \, du$$

 $\therefore C$

$$9 \quad v^2 = 2 \ln(3 + \cos x)$$

for stationary $v=0$

$$\therefore 0 = 2 \ln(3 + \cos x)$$

$$0 = \ln(3 + \cos x)$$

$$e^0 = 3 + \cos x$$

$$1 = 3 + \cos x$$

$$-2 = \cos x$$

no solⁿ $\therefore$ does not stop $\therefore A$

10

B

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Course		Name of task/exam	

Q11

$$a) m_1 = \frac{1}{2}$$

$$m_2 = 3$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{\frac{1}{2} - 3}{1 + \frac{3}{2}} \right|$$

$$= \left| \frac{-\frac{5}{2}}{\frac{5}{2}} \right|$$

$$= |-1|$$

$$\tan \theta = 1$$

$$\therefore \theta = \frac{\pi}{4}$$

$$b) \int \frac{1}{25+9x^2} dx$$

$$= \frac{1}{15} \tan^{-1} \frac{3x}{5} + C$$

$$c) \frac{1}{\sqrt{1-4x^6}} \times 6x^2$$

$$= \frac{6x^2}{\sqrt{1-4x^6}}$$

$$d) P(x) = x^3 - 3x^2 + kx + 12$$

Let roots be $\alpha, -\alpha, \beta$

sum of roots 1 at a time

$$\beta = 3$$

sum of roots 2 at a time:

$$-\alpha^2 + \alpha\beta - \alpha\beta = k$$

$$-\alpha^2 = k$$

$$-(\alpha^2) = k$$

product of roots:

$$-\alpha^2\beta = -12$$

$$\alpha^2\beta = 12$$

$$\alpha^2(3) = 12$$

$$\alpha^2 = 4$$

$$\therefore \alpha = \pm 2$$

$$\therefore k = -(4)$$

$$k = -4$$

$$e) P(x) = x^3 - 3x + 6$$

$$P'(x) = 3x^2 - 3$$

$$\text{at } x=1$$

$$P'(1) = 3 - 3$$

$$= 0$$

$\therefore$ at $x=1$ there is a

stationary point.

Since Newton's method finds where the tangent at a point crosses the x -axis, this would not work for the tangent at $x=1$. It wouldn't cross the x -axis as it would be a horizontal tangent.

$\therefore$ Newton's method would not work at $x=1$. Page of

Solutions for exams and assessment tasks

Academic Year		Calendar Year	
Course		Name of task/exam	

$$f) \cot^2 \theta - \cot \theta = 1$$

$$\cot^2 \theta - \cot \theta - 1 = 0$$

$$\cot \theta = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-1)}}{2(1)}$$

$$= \frac{1 \pm \sqrt{1+4}}{2}$$

$$= \frac{1 \pm \sqrt{5}}{2}$$

since $0 < \theta < \frac{\pi}{2}$

in the first quadrant.

$\therefore$ all trig ratios are positive.

$$\therefore \cot \theta = \frac{1 + \sqrt{5}}{2}$$

RTS

$$\therefore \cot 2\theta = \frac{1}{2}$$

$$\text{LHS} = \cot 2\theta$$

$$= \frac{1}{\tan 2\theta}$$

$$= \frac{1}{\frac{2 \tan \theta}{1 - \tan^2 \theta}}$$

$$= \frac{1 - \tan^2 \theta}{2 \tan \theta}$$

$$\text{as } \cot \theta = \frac{1 + \sqrt{5}}{2}$$

$$\tan \theta = \frac{2}{1 + \sqrt{5}}$$

$$\left(\text{since } \tan \theta = \frac{1}{\cot \theta} \right)$$

$$\therefore \tan \theta = \frac{2}{1 + \sqrt{5}}$$

$$\text{LHS} = 1 - \frac{4}{(1 + \sqrt{5})^2}$$

$$2 \left(\frac{2}{1 + \sqrt{5}} \right)$$

$$= \frac{(1 + \sqrt{5})^2 - 4}{(1 + \sqrt{5})^2} \div \frac{4}{(1 + \sqrt{5})}$$

$$= \frac{1 + 2\sqrt{5} + 5 - 4}{(1 + \sqrt{5})^2} \times \frac{(1 + \sqrt{5})}{4}$$

$$= \frac{2 + 2\sqrt{5}}{4(1 + \sqrt{5})}$$

$$= \frac{2(1 + \sqrt{5})}{4(1 + \sqrt{5})}$$

$$= \frac{2}{4}$$

$$= \frac{1}{2}$$

$$= \text{RHS}$$

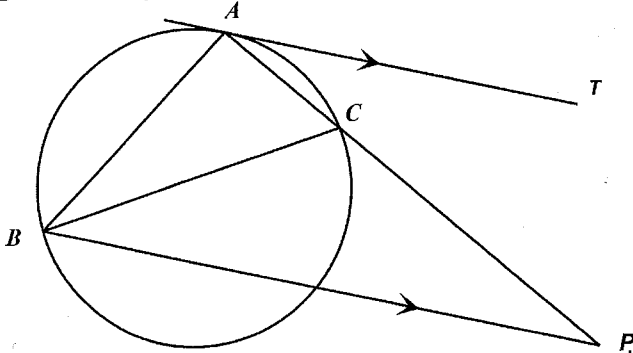
$\therefore$ shown

Solutions for exams and assessment tasks

Academic Year		Calendar Year	
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Q12

a.

AT is a tangent and is parallel to BP. Prove $\angle ABP = \angle ACB$.Let $\angle ABC = x$

$\therefore \angle TAC = x$ (angle between a tangent and a chord equals the angle in the alternate segment).

$\angle TAC = \angle CPB = x$ (alternate angles equal $AT \parallel BP$).

Let $\angle CBP = y$.

$\therefore \angle ABP = x + y$ (adjacent angles)

$\angle ACB = x + y$ (exterior angle of $\triangle CBP$ equals sum of 2 interior opposite angles).

$\therefore \angle ABP = \angle ACB$.

b. i. $T = 23 + Ae^{-kt}$ ①

$$\frac{dT}{dt} = -kAe^{-kt}$$

from ① $Ae^{-kt} = T - 23$

$$\therefore \frac{dT}{dt} = -k(T - 23)$$

ii. $T = 23 + Ae^{-kt}$

when $t = 0, T = 75$

$$75 = 23 + Ae^0$$

$$A = 75 - 23 = 52$$

iii. $t = 5, T = 65$

$$65 = 23 + 52e^{-k \times 5}$$

$$42 = 52e^{-5k}$$

$$\frac{42}{52} = e^{-5k}$$

$$\ln\left(\frac{42}{52}\right) = -5k$$

$$k = -\frac{1}{5} \ln\left(\frac{42}{52}\right)$$

iv. $T = ? \quad t = 8$

$$T = 23 + 52e^{-\left[-\frac{1}{5} \ln\left(\frac{42}{52}\right)\right] \times 8}$$

$$T = 60^\circ$$

$$c. \int \frac{e^x dx}{\sqrt{1-e^{2x}}}$$

$$u = e^x$$

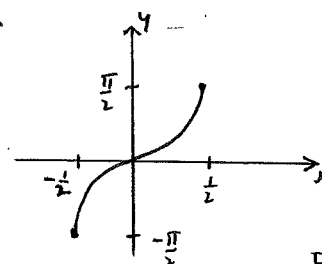
$$du = e^x dx$$

$$= \int \frac{du}{\sqrt{1-u^2}}$$

$$= \sin^{-1} u + C$$

$$= \sin^{-1} e^x + C$$

d. ii



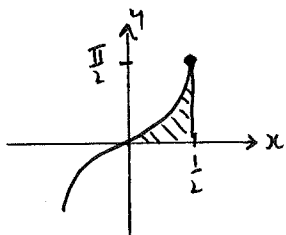
i. D: $-1 \leq x \leq 1$
 $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

R: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

Solutions for exams and assessment tasks

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iii



$V = \text{cylinder} - V_{\text{curve } y\text{-axis.}}$

$$V = \pi \left(\frac{1}{2}\right)^2 \frac{\pi}{2} - \int_0^{\frac{\pi}{2}} \pi x^2 dy$$

$$= \frac{\pi^2}{8} - \pi \int_0^{\frac{\pi}{2}} x^2 dy$$

Now $y = \sin^{-1} 2x$

$$\sin y = 2x$$

$$x = \frac{1}{2} \sin y$$

$$\therefore V = \frac{\pi^2}{8} - \pi \int_0^{\frac{\pi}{2}} \left(\frac{1}{2} \sin y\right)^2 dy$$

$$= \frac{\pi^2}{8} - \frac{\pi}{4} \int_0^{\frac{\pi}{2}} \sin^2 y dy$$

$$= \frac{\pi^2}{8} - \frac{\pi}{4} \int_0^{\frac{\pi}{2}} \left(\frac{1}{2} - \frac{1}{2} \cos 2y\right) dy$$

$\cos 2y = 1 - 2\sin^2 y$
 $2\sin^2 y = 1 - \cos 2y$
 $\sin^2 y = \frac{1}{2} - \frac{1}{2} \cos 2y$

$$= \frac{\pi^2}{8} - \frac{\pi}{4} \left[\frac{1}{2} y - \frac{\sin 2y}{4} \right]_0^{\frac{\pi}{2}}$$

$$= \frac{\pi^2}{8} - \frac{\pi}{4} \left[\left(\frac{\pi}{4} - 0\right) - (0 - 0) \right]$$

$$= \frac{\pi^2}{8} - \frac{\pi^2}{16}$$

$$= \frac{\pi^2}{16} \text{ units}^3$$

Q13

$$a. i. V^2 = 84 + 16x - 4x^2$$

$$\frac{1}{2} V^2 = 42 + 8x - 2x^2$$

$$\frac{d}{dx} \left(\frac{1}{2} V^2 \right) = \frac{d}{dx} (42 + 8x - 2x^2)$$

$$a = 8 - 4x$$

$$\therefore \ddot{x} = 8 - 4x$$

$$ii. \ddot{x} = -4(x - 2)$$

which is of the form

$$\ddot{x} = -n^2(x - h)$$

$\therefore$ SHM

$$iii. \therefore \text{Centre is at } x = 2.$$

when $v = 0$

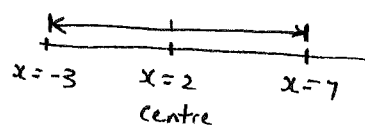
$$84 + 16x - 4x^2 = 0$$

$$21 + 4x - x^2 = 0$$

$$x^2 - 4x - 21 = 0$$

$$(x + 3)(x - 7) = 0$$

$$\therefore x = -3, x = 7.$$



$$\therefore \text{amplitude} = 5$$

$$\text{period} = \frac{2\pi}{\omega}$$

$$= \frac{2\pi}{2}$$

$$\therefore \text{Period} = \pi \text{ sec.}$$

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b) i $P(x) = (x-3)(x+1)^2(x-h)$

$(1, 0)$ satisfies

$$0 = (1-3)(1+1)^2(1-h)$$

$$0 = (-2)(4)(1-h)$$

$$1-h=0$$

$$h=1$$

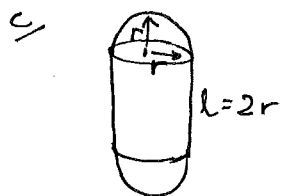
$$\therefore P(x) = (x-3)(x-1)(x+1)^2$$

ii $P(x) = x^4 - 2x^3 - 4x^2 + 2x + 3$

$$\frac{P(x)}{Q(x)} = \frac{x^4 - 2x^3 - 4x^2 + 2x + 3}{x^2 + 1}$$

$$\begin{array}{r} x^2 - 2x - 5 \\ x^2 + 1 \overline{) x^4 - 2x^3 - 4x^2 + 2x + 3} \\ \underline{x^4 + x^2} \\ -2x^3 - 5x^2 + 2x + 3 \\ \underline{-2x^3 - 2x} \\ -5x^2 + 4x + 3 \\ \underline{-5x^2 - 5} \\ 4x + 8 \end{array}$$

$\therefore$ remainder is $4x+8$



$V = \text{cylinder} + 2 \text{ hemi-spheres}$

$$V = \pi r^2 h + \frac{4}{3} \pi r^3$$

$$V = \pi r^2 (2r) + \frac{4}{3} \pi r^3$$

$$V = 2\pi r^3 + \frac{4}{3} \pi r^3$$

$$V = 3\frac{1}{3} \pi r^3$$

$$V = \frac{10}{3} \pi r^3$$

$$\frac{dV}{dr} = 10\pi r^2$$

$$\frac{dV}{dt} = 10 \text{ cm}^3/\text{s}$$

$$\therefore \frac{dr}{dt} = \frac{dr}{dV} \times \frac{dV}{dt}$$

$$= \frac{1}{10\pi r^2} \times 10$$

$$= \frac{1}{\pi r^2}$$

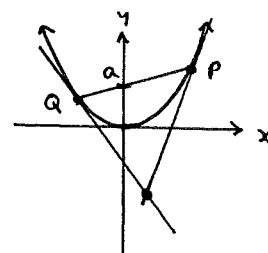
when $r=8$

$$\frac{dr}{dt} = \frac{1}{64\pi} \text{ cm/sec.}$$

d $x^2 = 4ay$

$P(2ap, ap^2)$

$Q(2aq, aq^2)$



i $m = \frac{aq^2 - ap^2}{2aq - 2ap}$

$$= \frac{a(q-p)(q+p)}{2a(q-p)}$$

$$= \frac{p+q}{2}$$

eqn $y - ap^2 = \frac{p+q}{2}(x - 2ap)$

$$2y - 2ap^2 = (p+q)x - (p+q)(2ap)$$

$$2y - 2ap^2 = (p+q)x - 2ap^2 - 2apq$$

$$(p+q)x - 2y - 2apq = 0$$

ii If focal chord $(0, a)$ satisfies

$$0 - 2a - 2apq = 0$$

$$2a = -2apq$$

$$-1 = pq$$

$$\therefore pq = -1$$

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iii $y = \frac{x^2}{4a}$

$$\frac{dy}{dx} = \frac{2x}{4a} = \frac{x}{2a}$$

at $P(2ap, ap^2)$

$$m_{\text{tang}} = \frac{2ap}{2a} = p$$

$\therefore$ eqn

$$y - ap^2 = p(x - 2ap)$$

Similarly eqn tangent at Q is

$$y - aq^2 = q(x - 2aq)$$

Solving simultaneously:

$$p(x - 2ap) + ap^2 = q(x - 2aq) + aq^2$$

$$px - 2ap^2 + ap^2 = qx - 2aq^2 + aq^2$$

$$px - qx = ap^2 - aq^2$$

$$x(p - q) = a(p - q)(p + q), \quad p \neq q$$

$$x = a(p + q)$$

$$y = px - 2ap^2 + ap^2$$

$$= p[a(p + q)] - ap^2$$

$$= ap^2 + apq - ap^2$$

$$= apq$$

$$\therefore x = a(p + q) \quad y = apq$$

since $pq = -1$

$$y = -a$$

$\therefore$ locus is $y = -a$

Q14

a RTP

$$\sum_{r=1}^n \ln\left(\frac{r+1}{r}\right) = \ln(n+1)$$

Step 1: Prove true for $n=1$

$$\text{LHS} = \ln 2$$

$$\text{RHS} = \ln(1+1) = \ln 2$$

$\therefore$ true for $n=1$

Step 2: Assume true for $n=k$

i.e.

$$\ln 2 + \ln\left(\frac{3}{2}\right) + \dots + \ln\left(\frac{k+1}{k}\right) = \ln(k+1)$$

Step 3: Prove true for $n=k+1$

i.e.

$$\ln 2 + \ln\left(\frac{3}{2}\right) + \dots + \ln\left(\frac{k+1}{k}\right) + \ln\left(\frac{k+2}{k+1}\right) = \ln(k+2)$$

$$\text{LHS} = \underbrace{\ln 2 + \ln\left(\frac{3}{2}\right) + \dots + \ln\left(\frac{k+1}{k}\right)}_{\ln(k+1)} + \ln\left(\frac{k+2}{k+1}\right)$$

$$= \ln(k+1) + \ln\left(\frac{k+2}{k+1}\right) \quad \text{by assumption}$$

$$= \ln(k+1) + \ln(k+2) - \ln(k+1)$$

$$= \ln(k+2)$$

$$= \text{RHS}$$

$\therefore$ prove by M.I. for all positive integers n .

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$$b) \quad 2 \cos x = \sqrt{3} \cot x$$

$$2 \cos x = \sqrt{3} \frac{\cos x}{\sin x}$$

$$2 \cos x - \frac{\sqrt{3} \cos x}{\sin x} = 0$$

$$\cos x \left(2 - \frac{\sqrt{3}}{\sin x} \right) = 0$$

$$\therefore \cos x = 0 \quad 2 = \frac{\sqrt{3}}{\sin x}$$

$$\sin x = \frac{\sqrt{3}}{2}$$

$$\therefore x = k\pi + \frac{\pi}{2}, \quad k \text{ integer}$$

and

$$x = k\pi + (-1)^k \frac{\pi}{3}, \quad k \text{ integer}$$

$$c) \quad i) \quad \ddot{y} = -g$$

$$\dot{y} = \int -g dt$$

$$\dot{y} = -gt + c_1$$

$$\text{when } t = 0 \quad \dot{y} = 0 \Rightarrow c_1 = 0$$

$$\dot{y} = -gt$$

$$y = \int -gt dt$$

$$y = -\frac{gt^2}{2} + c_2$$

$$\text{when } t = 0 \quad y = 40$$

$$c_2 = 40$$

$$\therefore y = -\frac{1}{2}gt^2 + 40$$

$$\text{when } g = 10$$

$$y = -5t^2 + 40$$

$$\ddot{x} = 0$$

$$\dot{x} = \int 0 dt$$

$$\dot{x} = c_3$$

$$\text{when } t = 0 \quad \dot{x} = 40 \cos 0 = 40$$

$$\therefore c_3 = 40$$

$$\dot{x} = 40$$

$$x = \int 40 dt$$

$$x = 40t + c_4$$

$$\text{when } t = 0 \quad x = 0 \Rightarrow c_4 = 0$$

$$\therefore x = 40t$$

$$ii) \quad \text{object hits when } y = 0$$

$$\therefore 0 = -5t^2 + 40$$

$$5t^2 = 40$$

$$t^2 = 8$$

$$t = 2\sqrt{2} \quad (t > 0)$$

$$\text{and } x = 40 \times 2\sqrt{2} = 80\sqrt{2} \text{ m.}$$

$$iii) \quad v = \sqrt{\dot{x}^2 + \dot{y}^2}$$

$$= \sqrt{40^2 + (-10\sqrt{8})^2}$$

$$= 48.99 \text{ m/s}$$

$$\tan \theta = \frac{\dot{y}}{\dot{x}}$$

$$\tan \theta = \frac{10\sqrt{8}}{40}$$

$$\theta = 35^\circ 16'$$

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$$d \quad (1-x)^n \left(1 + \frac{1}{x}\right)^n$$

$$= \left[(1-x)\left(1 + \frac{1}{x}\right)\right]^n$$

$$= \left[1 + \frac{1}{x} - x - 1\right]^n$$

$$= \left[\frac{1}{x} - x\right]^n$$

$$= {}^nC_0 \left(\frac{1}{x}\right)^n (-x)^0 + {}^nC_1 \left(\frac{1}{x}\right)^{n-1} (-x) + {}^nC_2 \left(\frac{1}{x}\right)^{n-2} (-x)^2 + \dots$$

Consider $(1-x)^n = {}^nC_0 - {}^nC_1 x + {}^nC_2 x^2 - {}^nC_3 x^3 + \dots + (-1)^n {}^nC_n x^n$

$$\left(1 + \frac{1}{x}\right)^n = {}^nC_0 + {}^nC_1 x^{-1} + {}^nC_2 x^{-2} + {}^nC_3 x^{-3} + \dots + {}^nC_n x^{-n}$$

$$(1-x)^n \left(1 + \frac{1}{x}\right)^n = {}^nC_0 ({}^nC_0 + {}^nC_1 x^{-1} + \dots) - {}^nC_1 ({}^nC_0 + {}^nC_1 + \dots) + \dots$$

Coeffs of x^2 term:

$${}^nC_2 {}^nC_0 - {}^nC_3 {}^nC_1 + {}^nC_4 {}^nC_2 - {}^nC_5 {}^nC_3 + \dots + (-1)^n {}^nC_n {}^nC_{n-2}$$

Coeff of x^2 term in $\left(\frac{1}{x} - x\right)^n$

$$T_{k+1} = {}^nC_k \left(\frac{1}{x}\right)^{n-k} (-x)^k$$

$$= {}^nC_k x^{k-n} (-1)^k x^k$$

$$= {}^nC_k x^{2k-n} (-1)^k$$

if x^2 term

$$2k - n = 2$$

$$2k = 2 + n$$

$$k = \frac{2+n}{2}$$

$\therefore n$ must be even.

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$\therefore$ for even n ,

$$\binom{n}{2}\binom{n}{0} - \binom{n}{3}\binom{n}{1} + \dots + (-1)^n \binom{n}{n}\binom{n}{n-2} = \binom{n}{\frac{n+2}{2}}(-1)^{\frac{n+2}{2}}$$

for odd n ,

$$\binom{n}{2}\binom{n}{0} - \binom{n}{3}\binom{n}{1} + \dots + (-1)^n \binom{n}{n}\binom{n}{n-2} = 0.$$